Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/01/2005
PI Name:	Fritzsch, Bernd Ph.D.		
Project Title:	Neurobiology Of Vestibular Development In Mutant Mice		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:	NOTE (June 2016): Dr. Fritzsch is currently at University of Iowa		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Caprona, Dominique (Creighton University) Beisel, Kirk W (Creighton University) Nichols, Dave H (Creighton University)		
Grant/Contract No.:	NAG2-1595		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To investigate the effect of normal and hypergravity on vestibular development in mice by investigating the effect of absence of otoconia on gravistatic behavior, neuronal connections and sensory epithelia gene expression. Using otoconia mutant mice we can investigate the absence of linear acceleration perception such as gravity on the behavior and connectional development of mice raised in 1g. Such genetically controlled manipulations allow for a more standardized analysis of a greater number of animals. By understanding the effect of absence of linear acceleration perception (gravity) on the functional development of ear input into the behavior and neuronal organization, we will be able to provide a more molecular understanding of the anticipated changes in the absence of gravity.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our results from a study of embryonic rats developing in microgravity as well as several studies using hypergravity exposure suggest that exposure to altered gravity alters the development of gravistatic vestibular connections and causes long lasting behavioral changes. We now propose to examine how vestibular connections are affected by hypergravity and whether there is a critical age during which gravistatic vestibular connections must be exposed to 1g in order to develop normal connections. The connections of specific inner ear receptors to their central nuclei will be analyzed using the lipophilic dye, DiI, to trace fibers in aldehyde fixed tissue and to analyze their connections to vestibular neurons both light and electron microscopically. In a first, descriptive approach, we will examine the normal sequence of development and how it is altered by exposure to hypergravity by exposing developing rat embryos and pups to hypergravity (2g) for variable periods of time before and after birth. Subsequently we will seek a possible critical age during which the formation of abnormal connections between the ear and brain caused by hypergravity can be appropriately reorganized by intermittent exposure to 1g. For this we will expose rat pups to hypergravity with intermittent exposures to 1g to counteract the hypergravity-induced effects. It is hoped that this program will ultimately help to identify the ages during which biologically relevant gravity stimuli are essential for normal vestibular development. Thus it may be possible to minimize long-term effects of microgravity on the development of vestibular connections during long-term flights by using a 1g centrifuge at appropriate intervals.
Task Progress:	We have made good progress on the behavioral studies and have recently pbulished a paper (de Caprona MD, Beisel KW, Nichols DH, Fritzsch B. 2004. Partial behavioral compensation is revealed in balance tasked mutant mice lacking otoconia. Brain Res Bull 64: 289-301). We are currently investigating the detailed projections of various endorgans in wildtype and tlt mutant mice and hope to submit another paper by 9/2005
Bibliography Type:	Description: (Last Updated: 03/26/2008)
Articles in Peer-reviewed Journals	de Caprona, M.D.; Beisel, K.W.; Nichols, D.H.; Fritzsch, B.; "Partial behavioral compensation is revealed in balance tasked mutant mice lacking otoconia." Brain Res Bull 64: 289-301 , Dec-2004